

Discrete convolution

For input X and kernel K

$$X = \begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{14} \\ x_{21} & x_{22} & x_{23} & x_{24} \\ x_{31} & x_{32} & x_{33} & x_{34} \\ x_{41} & x_{42} & x_{43} & x_{44} \end{bmatrix}, \quad K = \begin{bmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \\ k_{31} & k_{32} & k_{33} \end{bmatrix}.$$

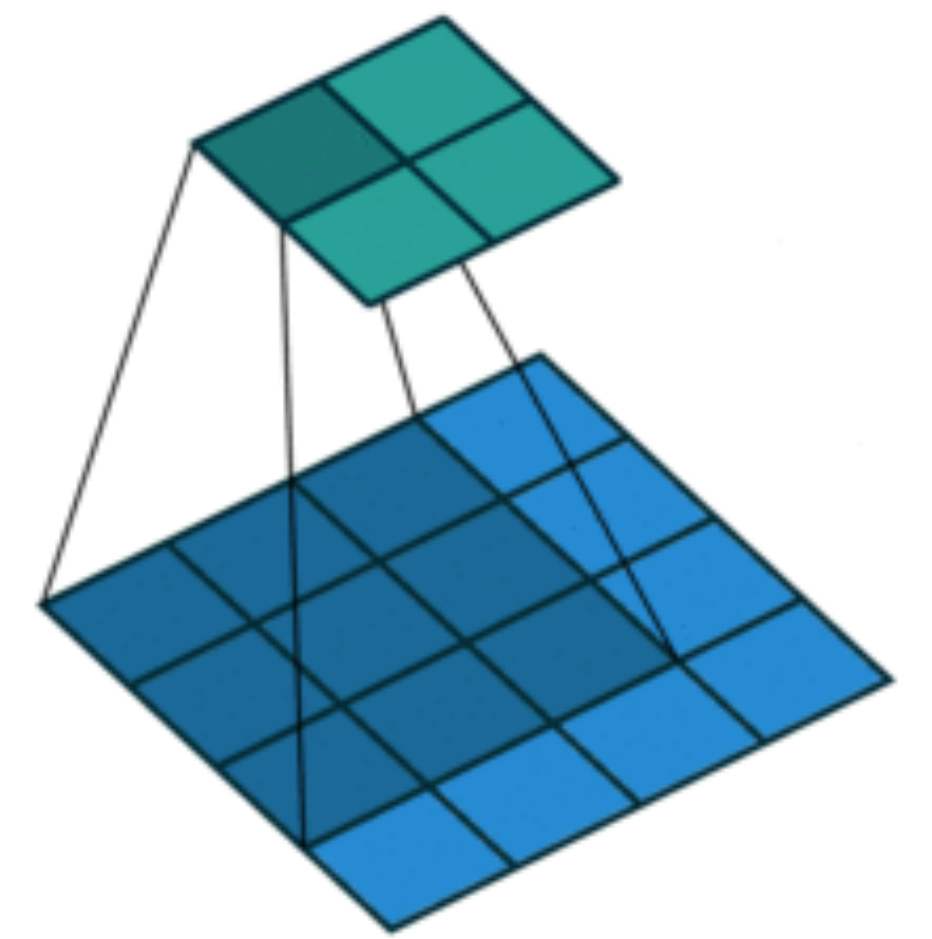
our output size with no padding and stride 1 will be:

$$(4 - 3 + 1) \times (4 - 3 + 1) = 2 \times 2$$

the general formula for the convolution is:

$$Y_{ij} = \sum_{m=1}^3 \sum_{n=1}^3 X_{i+m-1, j+n-1} \cdot K_{mn}.$$

Step by step



- Position (1,1), kernel aligned with top-left 3x3 patch of input

$$Y_{11} = x_{11}k_{11} + x_{12}k_{12} + x_{13}k_{13} + x_{21}k_{21} + x_{22}k_{22} + x_{23}k_{23} + x_{31}k_{31} + x_{32}k_{32} + x_{33}k_{33}$$

- Position (1,2), slide one column right

$$Y_{12} = x_{12}k_{11} + x_{13}k_{12} + x_{14}k_{13} + x_{22}k_{21} + x_{23}k_{22} + x_{24}k_{23} + x_{32}k_{31} + x_{33}k_{32} + x_{34}k_{33}$$

- Position (2,1), left edge, one row down

$$Y_{21} = x_{21}k_{11} + x_{22}k_{12} + x_{23}k_{13} + x_{31}k_{21} + x_{32}k_{22} + x_{33}k_{23} + x_{41}k_{31} + x_{42}k_{32} + x_{43}k_{33}$$

- Position (2,2), right edge, one row down

$$Y_{22} = x_{22}k_{11} + x_{23}k_{12} + x_{24}k_{13} + x_{32}k_{21} + x_{33}k_{22} + x_{34}k_{23} + x_{42}k_{31} + x_{43}k_{32} + x_{44}k_{33}$$